

P5255**TE/S/Insem. - 606****T.E. (Mechanical)****DESIGN OF MACHINE ELEMENTS -I****(2015 Course) (Semester - I) (302041)****Solution & Marking Scheme**

Q1

- a. The magnitude of factor of safety depends upon the following factors:

Effect of failure, Type of load, Degree of accuracy in force analysis, Material of the component, Reliability of the component, Cost of the component, Testing of machine element, Service conditions, Quality of manufacture.

(Anyeight : 04 Marks)

Q1
b

Given data.

$$P = 25 \text{ kN} = 25 \times 10^3 \text{ N} \quad \frac{b}{t} = \frac{2}{1} \quad S_{ut} = 400 \text{ N/mm}^2$$

$$f_s = 4.$$

① Permissible stresses

$$\sigma_{\max} = \frac{S_{ut}}{f_s} = \frac{400}{4} = 100 \text{ N/mm}^2$$

$$\sigma_t = \sigma_{\text{permissible}} = \sigma_{\max} = 100 \text{ N/mm}^2$$

Stress due to direct load .i.e direct stress

$$\sigma_d = \frac{P}{A} = \frac{25 \times 10^3}{b \cdot t} = \frac{25 \times 10^3}{2t^2} = \frac{12.5 \times 10^3}{t^2}$$

P.T.O.

② Bending Moment [M_b] [2 marks]

$$M_b = 25 \times 10^3 \times 140 = 3.5 \times 10^6 \text{ N-mm}$$

$$I = \frac{t \cdot b^3}{12} = \frac{1}{12} t \cdot (2t)^3 = \frac{2}{3} t^4$$

$$Y = \frac{b}{2} = \frac{2t}{2} = t \quad \dots [2 \text{ marks}]$$

③ Bending Stress [σ_b]

$$\sigma_b = \frac{M_b \cdot Y}{I} = \frac{3.5 \times 10^6 \times t}{\frac{2}{3} t^4} = \frac{5.25 \times 10^6}{t^3}$$

$$\sigma_{\max} = \sigma_D + \sigma_b$$

$$100 = \frac{12.5 \times 10^3}{t^2} + \frac{5.25 \times 10^6}{t^3}; \text{ Solving, we get}$$

$$t = 38.5 \text{ mm}$$

$$b = 77 \text{ mm}$$

..... [2 marks]

Q2 Given Data

$$F = 7.5 \text{ kN} = 7.5 \times 10^3 \text{ N} \quad \text{Short arm} = 100 \text{ mm}$$

$$\text{Long arm} = 500 \text{ mm} \quad S_{yt} = 300 \text{ N/mm}^2 \quad f_s = 5$$

$$\frac{b}{d} = \frac{3}{1} \text{ for pin}$$

$$\text{Pressure} = p = 10 \text{ N/mm}^2$$

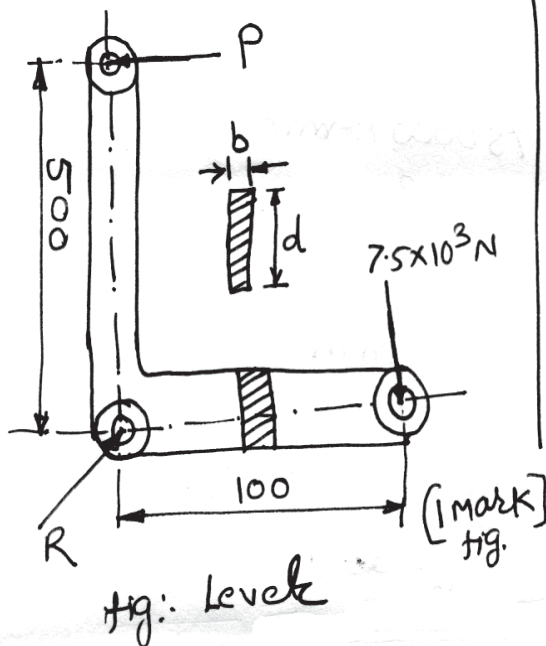
$$\text{Length to dia. ratio} = 1.5 : 1$$

(l) (d)

① Permissible stresses for pin & lever

$$\sigma_t = \frac{300}{5} = 60 \text{ N/mm}^2 \quad \dots\dots [1 \text{ mark}]$$

$$\tau = \frac{S_{sy}}{f_s} = \frac{0.5 \times 300}{5} = 30 \text{ N/mm}^2$$



Forces acting on the lever.

Taking moment about the axis of fulcrum

$$7.5 \times 10^3 \times 100 = P \times 500$$

$$\Rightarrow \boxed{P = 1500 \text{ N}}$$

Magnitude of ~~the~~ 'R' is

$$R = \sqrt{(7500)^2 + (1500)^2} = 7648.53 \text{ N} \quad [2 \text{ Marks}]$$

② Diameter and length of fulcrum pin.

$$R = P \times \text{Projected area of pin} = P \times (d_1 \times l_1)$$

$d_1 = \text{fulcrum pin diameter}$ & $l_1 = \text{fulcrum pin length}$ (mm)

$$7648.53 = 10 \times (d_1 \times 1.5 d_1)$$

$$\Rightarrow \boxed{d_1 = 22.581 \text{ mm}}$$

$$d_1 = 1.5 d_1 = 1.5 \times 22.58$$

$$d_1 = 33.87 \text{ mm}$$

[2 Marks]

③ Shear stress in the pin

Pin is subjected to double shear

$$\therefore \tau = \frac{R}{2 \times \left[\frac{\pi}{4} d_1^2 \right]} = \frac{7648.53}{2 \times \left[\frac{\pi}{4} \times (22.58)^2 \right]}$$

$$\tau = 9.55 \text{ N/mm}^2$$

[2 Marks]

④ using $d = 3b$

$$M_b = 7.5 \times 10^3 \times 100 = 750000 \text{ N-mm}$$

$$I = \frac{bd^3}{12} = \frac{1}{12} \times b \times (3b)^3 \text{ mm}^4$$

$$y = \frac{d}{2} = \frac{3b}{2} = 1.5b \text{ mm}$$

$$\sigma_b = \frac{M_b \times y}{I} \Rightarrow 60 = \frac{750000 \times 1.5b}{\frac{1}{12} \times b \times (3b)^3}$$

$$60 = \frac{750000 \times 1.5 \times 12}{27 b^3}$$

$$\Rightarrow \begin{cases} b = 20.27 \text{ mm} \\ d = 3b = 60.82 \text{ mm} \end{cases}$$

[2 Marks]

Q3

- a. For some machine tool applications, the shaft is designed on the basis of torsional rigidity. A transmission shaft is said to be torsionally rigid if it does not deflect too much under the action of external torque. Shaft design on the basis of torsional rigidity means design on the basis of permissible angle of twist per metre length of the shaft. The angle of twist in radians is given by

$$\theta_r = \frac{M_t l}{JG}$$

converting θ from radians to degrees

$$\theta = \frac{M_t l}{JG} \frac{180}{\pi} \text{ --- (a)}$$

for a solid circular shaft $J = \pi d^4/32$ – (b)

combining equations (a) and (b)

$$\theta = \frac{M_t l}{G} \frac{180}{\pi} (32/\pi d^4)$$

$$\theta = 584 M_t l / G d^4$$

θ = angle of twist (degrees)

l = length of shaft subjected to twisting moment (mm)

M_t = Torsional Moment or torque (N-mm)

G = Modulus of rigidity (N/mm²)

d = shaft diameter (mm)

Q3(b) Given data

$$n = 700 \text{ rpm} \quad P_m = 6.5 \text{ N/mm}^2 \quad \mu = 0.15$$

$$\text{length of hub} = l = 50 \text{ mm}$$

For splines

$$D = 40 \text{ mm}; d = 36 \text{ mm}; \text{No. of splines} = 08 = N$$

① length of hub

$$l = \frac{8 M_t}{P_m (D^2 - d^2) N} \quad [3 \text{ marks}]$$

$$50 = \frac{8 \times M_t}{6.5 \times 8 (40^2 - 36^2)}$$

$$M_t = 98800 \text{ N-mm}$$

② Torque transmitted

$$M_t = \frac{60 \times 10^6 [\text{KW}]}{2 \times \pi \times n}$$

$$98800 = \frac{60 \times 10^6 [\text{KW}]}{2 \times \pi \times 700}$$

$$KW = 7.24$$

[3 marks]

$$\text{Power transmitted} = 7.24 \text{ KW}$$

Q4 a

Flexible coupling

It can tolerate 0.5 mm of lateral axial misalignment and 1.5 degrees of angular misalignment

It prevents transmission of shock from one shaft to another and absorbs vibrations

It can be used for transmitting high torques

It is simple in construction and easy to assemble and dismantle

It requires more radial space

Rigid coupling has high torque transmitting capacity

Rigid coupling is easy to design and manufacture

Rigid coupling cannot tolerate misalignment between the axes of two shafts

Rigid coupling can be used only where the motion is free from shock and vibration

Rigid coupling also requires more radial space

Q4

(b) Given data.

$$D_o = 160 \text{ mm} \quad d_r = 95 \text{ mm} \quad N = 6$$

$$\text{Pre-load} = P_i = 10 \times 10^3 \text{ N} \quad \mu = 0.15 \quad n = 100 \text{ rpm}$$

① Preload in bolts (P_i)

$$P_i = \frac{M_t}{\mu N R_f}; \text{ where } R_f \text{ is friction radius}$$

$$R_f = \frac{2}{3} \frac{R_o^3 - R_i^3}{R_o^2 - R_i^2} \quad \text{As bolts are fitted in large clearance holes}$$

$$= \frac{2}{3} \frac{(80)^3 - (47.5)^3}{(80)^2 - (47.5)^2} = 65.1307 \text{ mm} \quad [2 \text{ Marks}]$$

② Torque transmitted

$$M_t = P_i \times \mu \times N \times R_f$$

$$= 10 \times 10^3 \times 0.15 \times 6 \times 65.1307$$

$$M_t = 586176 \text{ N-mm} \quad [2 \text{ Marks}]$$

③ Power transmitted by coupling

$$M_t = \frac{60 \times 10^6 [\text{KW}]}{2 \times \pi \times n} = \frac{60 \times 10^6 \times (\text{KW})}{2 \times \pi \times 100}$$

$$\text{KW} = 6.1384$$

$$\therefore \text{Power transmitted} = 6.138 \text{ KW}$$

[2 Marks]

Q5

(a) Endurance limit modifying factors

K_a = Surface finish factor

K_b = Size factor

K_c = Reliability factor

K_d = Modifying factor to account for stress concentration.

} ... [2 marks]

Explanation of any 2 of above [2 marks]
[1 mark each]

(b) Given data

$$d = 50 \text{ mm} \quad S_f = \sigma_b = 250 \text{ N/mm}^2$$

$$S_{ut} = 600 \text{ N/mm}^2 \quad R = 90\% \quad K_a = 0.44$$

$$K_b = 0.85 \quad K_c = 0.897$$

$$S_e = K_a K_b K_c S'_e$$

$$S'_e = 0.5 S_{ut} = 0.5 \times 600 = 300 \text{ N/mm}^2$$

$$S_e = 0.44 \times 0.85 \times 0.897 \times 300$$

$$\boxed{S_e = 100.64 \text{ N/mm}^2}$$

$$0.9 S_{ut} = 0.9 (600) = 540 \text{ N/mm}^2$$

$$\log_{10}(0.9 S_{ut}) = \log_{10}(540) = 2.7324$$

$$\log_{10}(S_e) = \log_{10}(100.64) = 2.0028$$

$$\log_{10}(S_f) = \log_{10}(250) = 2.3979$$

$$\log_{10}(10^3) = 3 \quad \& \quad \log_{10}(10^6) = 6$$

} [2 marks]

The S-N curve

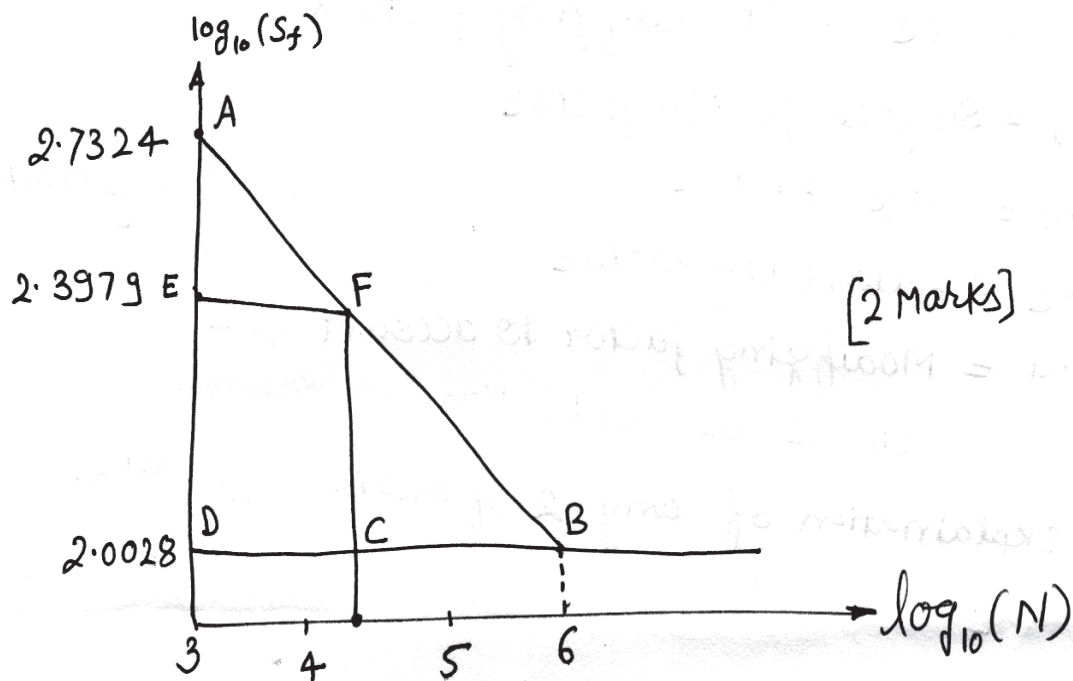


fig: S-N curve for steel bar

fatigue life of bar

$$\bar{EF} = \frac{\bar{DB} \times \bar{AE}}{\bar{AD}} = \frac{(6-3)(2.7324-2.3979)}{(2.7324-2.0028)}$$

$$\therefore \log_{10}(N) = 3 + 1.3754$$

$$\boxed{N = 23736.2 \text{ cycles}} \quad [2 \text{ Marks}]$$

Q6 (a) S-N curve for steels

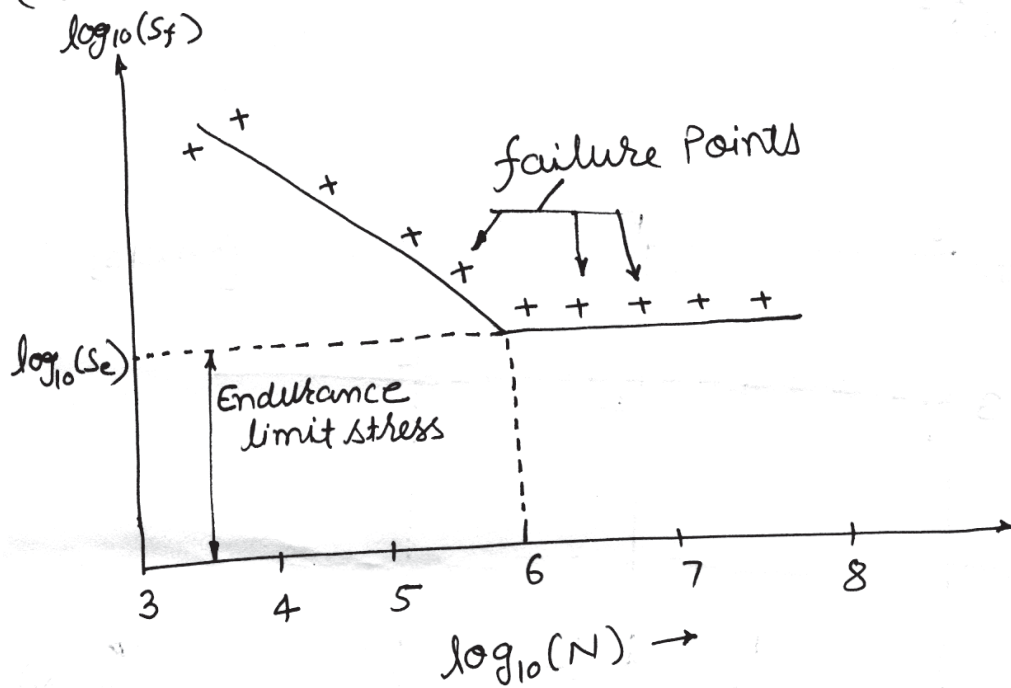


fig:- S-N curve for steels

Correct labelled diagram - 2 Marks
 Explanation - 2 Marks

(b) Given data

$$S_{ut} = 580 \text{ N/mm}^2 \quad S_e = 275 \text{ N/mm}^2$$

$$N = 90,000 \text{ cycles.}$$

$$0.9 S_{ut} = 0.9 \times 580 = 522 \text{ N/mm}^2$$

$$\log_{10}(0.9 S_{ut}) = 2.7176$$

$$\log_{10}(S_e) = \log_{10}(275) = 2.4393$$

$$\log_{10}(90,000) = 4.9542$$

$$\log_{10}(10^3) = 3 \text{ and } \log_{10}(10^6) = 6$$

2 [marks]

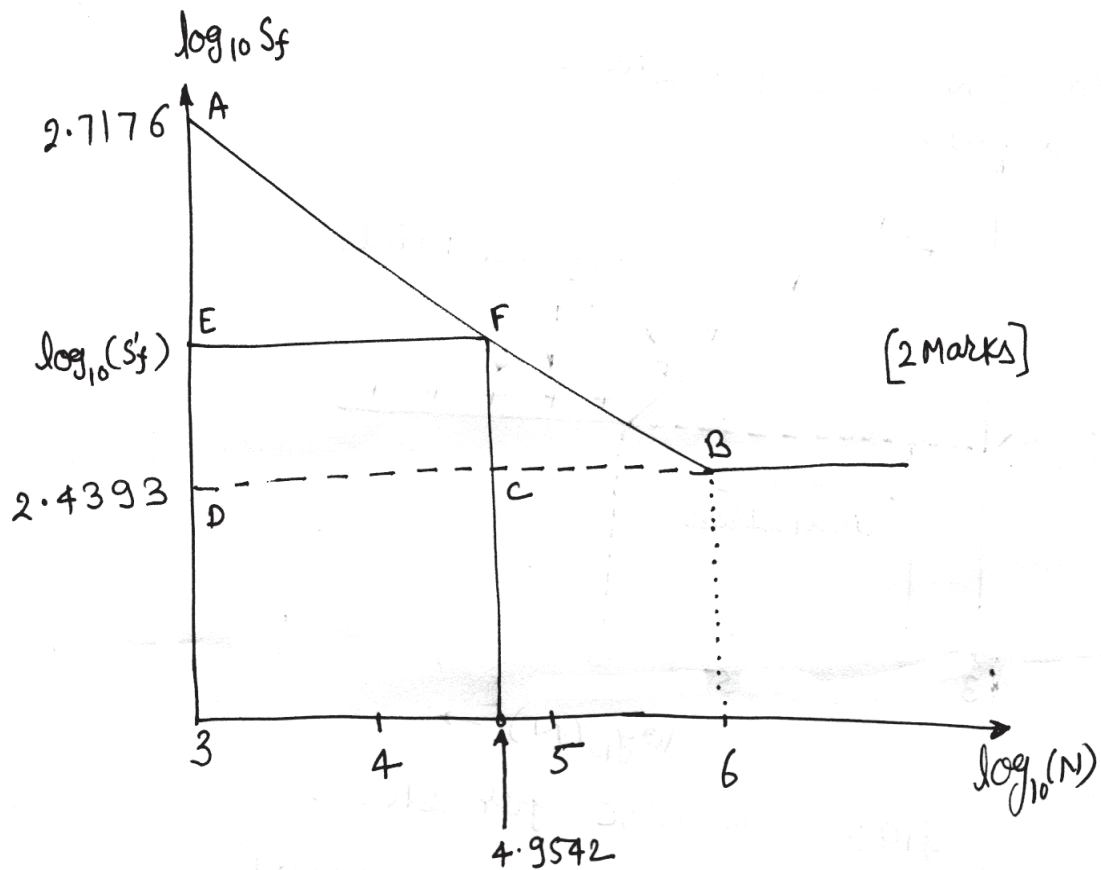


fig S-N. curve for rotating bar

$$\log_{10}(s'_f) = 2.7176 - \frac{(2.7176 - 2.4393)}{(6 - 3)} \times (4.9542 - 3)$$

i.e

$$\log_{10}(s'_f) = 2.7176 - \frac{(2.7176 - 2.4393)}{(6 - 3)} \times (4.9542 - 3)$$

$$\log_{10}(s'_f) = 2.5363$$

$$\Rightarrow \boxed{S'_f = 343.807 \text{ N/mm}^2} \quad [2 \text{ Marks}]$$

EEE